

Keywords: neural collapse • representation learning • information theory • language models

Neural Collapse Is Forbidden

An Information Floor Theorem Shows That Within-Class Variance in Language Models Is Obligatory, Not a Training Artifact

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arXiv:2607.09487 • July 10, 2026

Neural collapse—the phenomenon where final-layer representations within each class converge to a single point during training—has been hailed as evidence that deep networks find elegant, maximally-separated class geometries. But for language models, this paper proves that neural collapse is theoretically impossible: within-class variance cannot shrink below a positive lower bound determined by how much contextual information each token carries beyond its category label. The paper terms this limit the **information floor**, derives it analytically, and confirms it empirically across 14 models spanning a 100× parameter range.

AT A GLANCE

Problem: Within-class variance in LM representations is commonly attributed to incomplete training or sub-optimal regularization.

Why it matters: If within-class variance is obligatory, attempts to reduce it via regularization or architecture choices will hurt model performance.

Method: Probing study across 14 models; theoretical proof of an information floor on within-class variance; k-NN mutual information estimator to compute the bound empirically.

Result: Within-class variance accounts for 79–91% of total representational variance; the theoretical floor covers 71–79%.

Evidence: 26-page paper with 3 figures and 19 tables across five text domains and five model families.

The Big Picture

Neural collapse was first described in classification networks: as training progresses, the penultimate-layer representations of samples within each class collapse to a single class mean, and the class means arrange themselves in a maximally-equidistant “simplex ETF” geometry. The phenomenon has been viewed as a signature of an ideal training outcome—a fully collapsed network is maximally linear-separable and robust to perturbations.

For language models, the analogue of a “class” is ambigu-

ous. Prior work has used token categories (part of speech, semantic class, domain label) as the class variable and found that within-category variance is large—sometimes 80–90% of total variance—and interpreted this as evidence that language models fail to achieve neural collapse.

This paper argues the interpretation is exactly backwards: large within-class variance is not a failure but a necessity. Language models must encode token-specific contextual information within each category because two tokens of the same category (e.g., both nouns) carry different next-token distributions depending on their context. Collapsing representations within a category would destroy this information and increase prediction loss—so the training objective itself prevents neural collapse.

Why Existing Approaches Fall Short

The neural collapse literature for classification assumes a fixed label space where two samples of the same class carry identical distributional targets. This assumption fails for language models: the distribution over next tokens for “bank” depends on whether the preceding context is financial or geographic, even though “bank” is in the same syntactic/semantic category in both cases.

Attempts to fine-tune language models toward neural collapse (via additional regularization or loss terms) have been tried in the context of classification fine-tuning. The paper’s theoretical result explains why these attempts always succeed in reducing within-class variance on the fine-tuning data but increase perplexity on held-out data: they destroy the contextual information encoded in the within-class variance.

Core Mechanism

The information floor arises from the structure of next-token prediction. For a token X with category label C and context Z , the model must compute $p(X_{\text{next}} | X, Z)$, which depends on both the category and the specific context. Two tokens of the same category in different contexts have different conditional next-token distributions, and these differences must be encoded in the representation.

The conditional mutual information $I(X; Z | C)$ quantifies exactly how much contextual information is necessary beyond the category label. The paper proves that the within-class variance σ_W^2 in any representation that achieves training loss below the theoretical minimum must satisfy $\sigma_W^2 \geq c \cdot I(X; Z | C)$.

Technical Formulation

Let X be the token, C its macro-category, Z the context window, and $h(X, Z) \in \mathbb{R}^d$ the final-layer representation.

Define the within-class variance as:

$$\sigma_W^2 = \mathbb{E}_C [\text{Var}(h(X, Z) \mid C)]$$

Theorem (Information Floor): For any model that achieves cross-entropy loss $\mathcal{L} \leq H(X_{\text{next}} \mid C, Z)$ on the training distribution:

$$\sigma_W^2 \geq c \cdot I(X; Z \mid C)$$

where $c > 0$ depends only on the loss function curvature and projection constants of the final linear layer.

The proof proceeds by lower-bounding the Fisher information of the output distribution as a function of σ_W^2 , and showing that $\sigma_W^2 < c \cdot I(X; Z \mid C)$ implies the Fisher information is insufficient to support the observed training loss.

Learning or Inference Procedure

1. For each of the 14 models, collect final-layer representations on 50,000 tokens from five text domains.
2. Assign 32-category macro-category labels using WordNet + POS tagging.
3. Compute within-class and between-class variance in the top- k principal subspace of the full covariance matrix.
4. Estimate $I(X; Z \mid C)$ using a k-NN mutual information estimator (Kozachenko-Leonenko calibrated).
5. Compare observed $\sigma_W^2 / (\sigma_B^2 + \sigma_W^2)$ against the theoretical lower bound derived from $I(X; Z \mid C)$.

What the Guarantee Says

The information floor theorem provides a constructive lower bound: given an estimate of $I(X; Z \mid C)$ from data, the bound predicts the minimum observable within-class variance share. Across five domains and 14 models, the theoretical floor covers 71–79% of total variance, and the observed share is 79–91%—consistently above the floor by a margin of 8–12 percentage points, consistent with the floor being a lower bound rather than an equality.

Experimental Findings

Model family	$\sigma_W^2 / \sigma_{\text{total}}^2$	Floor
GPT series	$82.4\% \pm 3.1\%$	73.1%
LLaMA series	$84.7\% \pm 2.4\%$	75.8%
Mistral	$83.9\% \pm 2.8\%$	74.2%
Claude family	$79.3\% \pm 4.2\%$	71.0%
Gemma	$85.2\% \pm 1.9\%$	76.3%

The variance share is insensitive to scale within each family (max 2% change over $10\times$ parameter increase) and insensitive to tokenization (BPE vs. SentencePiece: 0.8% difference). All observations lie above the predicted floor.

Ablations and Interpretation

Random-label control: Replacing token category labels with random assignments increases σ_B^2 by 18% and σ_W^2 by 22%, confirming the variance decomposition is meaningful.

Layer depth: Within-class variance share increases from 55% at layer 1 to 84% at the final layer, suggesting the model progressively specializes representations toward context-specific rather than category-level structure.

Fine-tuning toward collapse: Fine-tuning on sentiment classification with an additional within-class variance penalty reduces σ_W^2 on the fine-tuning domain by 12% but increases perplexity by 8%, directly confirming that forcing collapse destroys necessary contextual information.

Rare tokens: The information floor is higher for rare tokens (those with unique distributional contexts), consistent with the finding that rare tokens are disproportionately memorized—they carry more $I(X; Z \mid C)$ and therefore require more within-class dispersion to represent correctly.

Reference

Bruno Abrahao. **Neural Collapse Is Forbidden: Information Floors in Language Models.** arXiv:2607.09487, July 10, 2026. NYU Shanghai. <https://arxiv.org/abs/2607.09487>